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Two-dimensional Czochralski growth of single-crystal MoS_2

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Batch production of single-crystal two-dimensional (2D) transition metal dichalcogenides is one prerequisite for the fabrication of next-generation integrated circuits. Contemporary strategies for the wafer-scale high-quality crystallinity of 2D materials centre on merging unidirectionally aligned, differently sized domains. However, an imperfectly merged area with a translational lattice brings about a high defect density and low device uniformity, which restricts the application of the 2D materials. Here we establish a liquid-to-solid crystallization in 2D space that can rapidly grow a centimetre-scale single-crystal MoS_2 domain with no grain boundaries. The large MoS_2 single crystal obtained shows superb uniformity and high quality with an ultra-low defect density. A statistical analysis of field effect transistors fabricated from the MoS_2 reveals a high device yield and minimal variation in mobility, positioning this FET as an advanced standard monolayer MoS_2 device. This 2D Czochralski method has implications for fabricating high-quality and scalable 2D semiconductor materials and devices.

Batch production of large-scale high-quality 2D transition metal dichalcogenides (TMDCs) presents a major challenge in the industrial manufacturing of 2D devices[1–4]. Recent studies have reported the successful growth of wafer-scale MoS_2 single crystals by an evolutionary chemical vapour deposition (CVD) to merge substantial micrometre-scale domains with unidirectional alignment. However, the multi-seed growth method suffers from translational grain boundaries induced by imperfect splicing[5–9]. Alternatively, the synthesis of a macroscopic wafer-scale monolayer single crystal from a single nucleus provides another feasible method[10]. Due to the high nucleation density and slow growth rate of CVD, the competition between domains on the substrate leads to only millimetre-scale TMDC domain growth[11–13]. Some progress has been made concerning large-scale single-crystalline graphene with the single-seed growth method by reducing the nucleation density and accelerating the growth rate[14,15]. Unfortunately, large-scale single-crystal TMDC monolayers have not been realized via a similar single-seed growth approach, due to the fact that the edge formation energies and nucleation barrier are much smaller than those of graphene, making it extremely difficult to suppress the nucleation density to a very low level[16,17].

Crystallization from a liquid precursor stands as an efficient means for large-scale single-crystal preparation in semiconductor manufacturing. For example, the Czochralski method enables the production of bulk single crystals involving a liquid–solid (L–S) transformation from a liquid source to single-crystal silicon ingots[18]. Liquid precursors, such as sodium molybdate or tungstate droplets, have been introduced to enable the growth of TMDCs[12,19] for their low melting point. However, limited by the small wetted area of the liquid precursor and high nucleation density on conventional substrates, only submillimetre-scale TMDC domains have been obtained by the L–S reaction[20,21]. Hence, establishing a large-scale 2D liquid precursor on a wettable substrate and reducing nucleation density are critical

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prerequisites for growing large-scale 2D TMDCs in a process that resembles the Czochralski process[12].

Herein we propose a 2D Czochralski (2DCZ) method for the reliable growth of centimetre-scale single-crystal MoS_2 domains. Specifically, a solid–liquid–solid process has been introduced to enable the successful transformation from polycrystal MoS_2 to single-crystal MoS_2. First, a large-scale 2D liquid precursor film is formed on the molten glass substrate through the etching reaction (between oxygen and the pre-deposited polycrystal MoS_2) and the annealing process. Then, large-area MoS_2 domains on the atomically smooth interface are obtained with an ultrafast sulfuration process. The grown MoS_2 domains demonstrated a scale of 1.5 cm with an ultra-low defect density of 2.9×10^{12} cm^{-2}. Moreover, the low nucleation density weakens the adhesion between the MoS_2 film and the substrate, facilitating an ultra-clean, fast and high-quality transfer process with the aid of deionized water. The average mobility of the field effect transistor (FET) arrays is measured at 55 cm^2 V^{-1} s^{-1} with a low variation of 15.9%. The short-channel FET achieved a high saturated current of 443.8 μA $μm^{-1}$. The best mobility of a FET achieved was 105.4 cm^2 V^{-1} s^{-1}. The high quality and uniformity of the MoS_2 film enable the FET arrays to exhibit an impressive electrical performance, thereby promoting the application of 2D semiconductors from devices to advanced integrated circuits.

Crystallization of centimetre-scale MoS_2 domains

The traditional Czochralski method is an effective approach for growing large-sized, high-quality single-crystal materials. However, applying the conventional Czochralski method for MoS_2 growth poses significant challenges due to the ultra-high melting point (2,375 °C) and difficulty in controlling the thickness. Therefore, it is necessary to shift the approach by lowering the melting point of the precursor and ultimately achieving a liquid–solid process confined to a single molecular layer. By using a low-melting-point precursor and employing a chemical process to facilitate the liquid–solid transition, we successfully developed a confined 2DCZ method.

The 2DCZ method was employed for the growth of a large-scale MoS_2 monolayer in an atmospheric pressure tube furnace with MoO_3 and sulfur as precursors (Supplementary Fig. 1a). In comparison to the traditional CVD process (Fig. 1a), which completes the deposition process at the vapour–solid interface through adsorption, random nucleation and splicing, our method prefabricated a 2D liquid film to achieve the in-plane crystallization for large-sized MoS_2 domain growth. First, by tuning the partial pressure of O_2 and sulfur vapour, the pre-deposition and etching of the Mo source were realized, forming a liquid precursor (Fig. 1b, stages I and II). Subsequently, a eutectic reaction and liquid–liquid phase separation occurred on a molten glass substrate, resulting in a stable and homogeneous 2D liquid precursor (Fig. 1b, stage III). Finally, a liquid crystallization process was triggered by sulfur vapour (Fig. 1b, stage IV). The chemical reaction process was mapped using a quaternary phase diagram (Fig. 1c). The arrows from MoO_3 to MoS_2, MoS_2 to MoO_3, MoO_3 to $Na_2Mo_2O_7$ and $Na_2Mo_2O_7$ to MoS_2 correspond to the pre-deposition, etching, eutectic reaction and liquid crystallization processes, respectively. The 2DCZ process is detailed later in this paper.

Significantly, two essential factors must be simultaneously achieved for large-scale 2D single domain growth: first, increasing the nucleation barrier to reduce the nucleation density, and second, reducing the diffusion barrier to accelerate the growth rate around the nuclei[22]. CVD on atomically smooth molten glass substrates can suppress nucleation[13]. However, the rate-determining step of CVD is based on the adsorption rate of vapour precursors, which limits the growth rate and achieves only 2.6 mm MoS_2 growth. Since our work also uses molten glass as a substrate, we use a 2D liquid precursor to reduce the diffusion barrier, rather than the vapour deposition process. This improves the size of the as-prepared monolayer single-crystal MoS_2 domains to 1.5 cm and splices them into a continuous film (Supplementary Fig. 1b,c). This domain is among the largest MoS_2 monolayer single-crystal domains reported[12,13,23–30]. Through a rapid supersaturated crystallization process, a growth rate of 75 μm s^{-1} was achieved (Supplementary Video 1), which is faster than previous works for 2D MoS_2 growth (Supplementary Table 1).

Our method significantly enhances domain size and accelerates growth rates compared with previously reported methods for MoS_2 growth (Fig. 1d). The sulfur vacancy density in the MoS_2 is controlled to 2.9×10^{12} cm^{-2} due to a continuous and uniform crystallization process in an excess sulfur atmosphere, significantly lower than in MoS_2 prepared by mechanical exfoliation, physical vapour deposition or CVD[2,31–33] (Fig. 1e). Here we focus on the growth size and rate of the individual domain, and thus the scale of the spliced area is relatively small.

Mechanism of 2DCZ method

In this study, we introduced a 2DCZ method aiming to expand the area for the 2D liquid precursor while curbing nucleation, using an atomically smooth molten glass substrate and an ultrafast crystallization rate. The generation of the 2D liquid precursor involves etching the pre-deposited MoS_2. The liquid crystallization process was observed through in situ imaging. A microtube furnace equipped with an observation window facilitated the recording of the 2DCZ process (Methods and Supplementary Fig. 2a). The in situ imaging experiment was conducted in eight distinct stages to correspond with the pre-deposition, etching, eutectic reaction, spreading and in-plane crystallization parts of the 2DCZ process.

Stages I and II were conducted at 500–1,000 °C in an Ar atmosphere. During the heating process, the MoO_3 powder on the glass surface melted and reacted with Na_2O in the glass substrate, leading to the formation of numerous $Na_2Mo_2O_7$ eutectic droplets on the glass surface (Fig. 2a). This formation of $Na_2Mo_2O_7$ prevents the rapid sublimation of MoO_3, ensuring an adequate Mo source for the subsequent steps.

Stage III was conducted at 1,000–1,100 °C in an Ar/S atmosphere. The initial sulfuration process is induced by a sulfur atmosphere to achieve the pre-deposition of polycrystal MoS_2. Due to nucleation events in high-contact-angle precursor droplets, MoS_2 grows vertically along a Mo-rich direction, resulting in high-density multilayer MoS_2 domains floating on the molten glass, as shown in the optical micrograph of Fig. 2b, stage III. The initial pre-deposition of polycrystalline MoS_2 is designed to ensure that the Mo source is uniformly distributed on the glass substrate surface before it melts, rather than being evaporated.

Stage IV was conducted at 1,100 °C in an Ar/O_2 atmosphere. Introducing O_2 into the tube results in a solid-to-liquid (S–L) process of pre-deposited MoS_2 on the molten glass, displayed in the optical micrograph in Fig. 2b, stage IV. The S–L process represents a chemical process involving etching and a eutectic reaction (as shown in equation (1)).

$$2MoS_2 + Na_2O + 7O_2 \rightarrow Na_2Mo_2O_7 + 4SO_2 \quad (1)$$

The eutectic reaction is confirmed by a group of controlled experiments. A small amount of MoO_3 powder is dispersed onto the surface of the glass and heated in the Ar/S atmosphere. The heating process is stopped at stage II (Fig. 2a), stage III (Fig. 2b), stage VI (Fig. 2c) and stage VIII (Fig. 2d) and cooled to room temperature within 3 min by a water-cooling system. The production on the surface of the glass is confirmed by a Raman test (Supplementary Fig. 2b).

Stages V and VI were conducted at 1,100–1,100 °C in an Ar/O_2 atmosphere. Molten glass decreases the contact angle of eutectic droplets, forming a 2D liquid film (Supplementary Video 2). High-resolution in situ imaging, shown in Fig. 2c, captures the interference patterns during the formation of the 2D liquid film. This observed pattern closely resembles the formation of Longmuir–Blodgett films, indicating the ultrathin property of the liquid film[34]. The molten glass substrate is more conducive to the spread of liquid precursors, thereby facilitating the growth of monolayer MoS_2.

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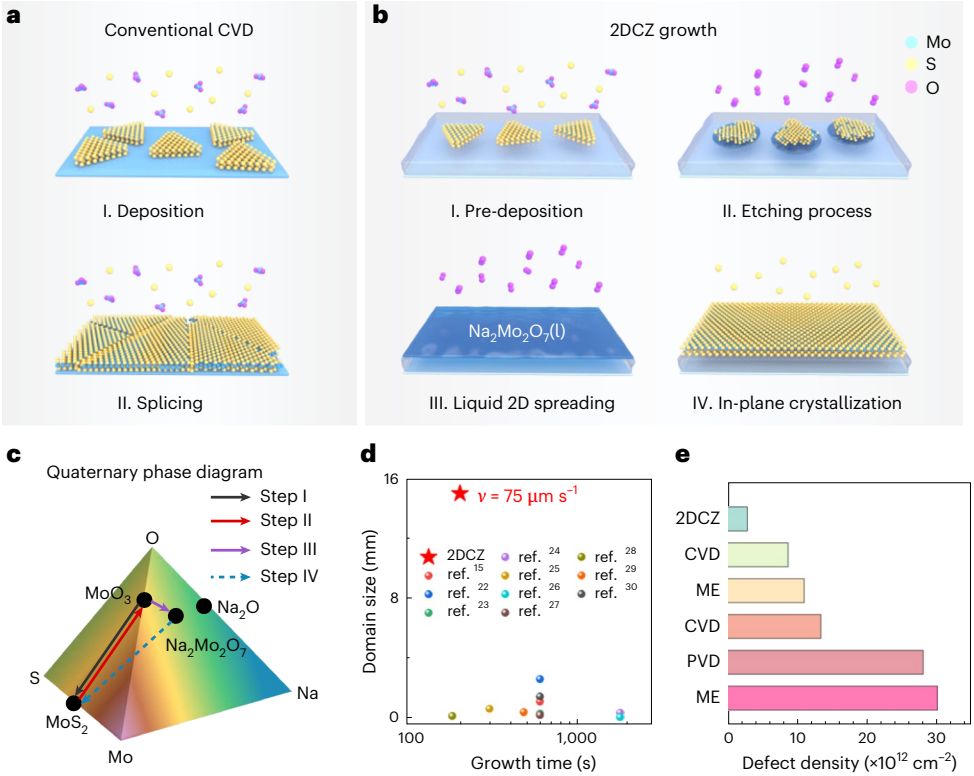

Fig. 1 | Crystallization of 2DCZ for large-scale, high-quality MoS_2 domain. **a**, Schematic of conventional CVD process for MoS_2 growth. **b**, Schematic overview of the 2DCZ growth process. I. Conventional V–S process for pre-deposition of MoS_2 domains. II. O_2-assisted etching process. III. The liquid eutectic precursor 2D film spreads and converges on the molten glass. IV. Large-scale 2DCZ growth under the sulfur atmosphere. **c**, Mo–O–S–Na quaternary phase diagram of 2DCZ: arrows from MoO_3 to MoS_2, MoS_2 to MoO_3, MoO_3 to $Na_2Mo_2O_7$ and $Na_2Mo_2O_7$ to MoS_2 illustrate the sulfuration, oxidation, eutectic reaction and in-plane crystallization processes, respectively. **d**, Benchmark comparison of the 2DCZ method: this shows the largest domain sizes achieved in various reports and their corresponding growth times. **e**, Comparison of defect density in MoS_2 achieved by mechanical exfoliation (ME), physical vapour deposition (PVD), CVD and 2DCZ.

Stages VII and VIII were conducted at 1,100–1,100 °C in an Ar/S atmosphere. After the spreading process, the O_2 supply is cut off, creating an excess sulfur vapour environment, enabling the achievement of large-sized MoS_2 domains (equation (2)). The MoS_2 domain with a side of length 750 μm is achieved in only 10 s, indicating a growth rate of 75 μm s^{-1} which is at the same level as the conventional Czochralski method[35] (Fig. 2d). The 2DCZ method achieves supersaturation and precipitation at the same temperature through a chemical reaction, thereby realizing the faster liquid-to-solid process than in the conventional CVD process for 2D materials growth.

$$Na_2Mo_2O_7 + 7S \rightarrow 2MoS_2 + Na_2O + 3SO_2 \quad (2)$$

To sum up, the 2DCZ process can be divided into two parts: the two-dimensionalization of the precursor and the rapid Czochralski process based on the two-dimensionalized precursor. Initially, we obtained a 2D supersaturated solution of MoS_2 through pre-deposition, etching, spreading and sulfurization. Subsequently, due to the environmental temperature being significantly lower than the melting point of MoS_2, single-crystal monolayer MoS_2 was precipitated through the Czochralski process.

Exploring reaction processes by analysing energy dynamics provides deep insights into the CVD growth mechanism[36] (Fig. 2e). Typically, during conventional CVD growth on a substrate surface, two energy barriers must be overcome, corresponding to the process of nucleation (ΔE_N) and diffusion (ΔE_D). According to the Kossel crystal model, during the vapour–solid (V–S) process of conventional CVD growth, active regions, such as defects, impurities, kinks and step edges (Fig. 2f), provide adsorption and nucleation sites due to their lowest formation energy and thus a low nucleation barrier[22]. However, while this process might lead to the formation of a 2D layer, it often results in grain boundaries that cannot be merged or stitched, leading to various challenges in application. Typically, the easier it is to nucleate on specific areas with lower energy, the higher the binding energy, resulting in a stronger adhesion to the substrate. Therefore, a large number of nucleation sites in the 2D layer would strengthen adhesion and make it challenging to separate a large-scale material from the substrate. Furthermore, the precursors of CVD need to be adsorbed on the substrate and to overcome a high diffusion barrier to lateral growth near the nucleation sites.

By contrast, the liquid phase introduced in our 2DCZ growth process significantly alters the reaction process. MoS_2 with large domain sizes is achieved by suppressing nucleation and promoting diffusion through the regulation of energy barriers. Initially, the molten glass exhibits an atomically smooth and defect-free surface, significantly enhancing the nucleation barrier[37,38]. This results in a low-density, uniformly distributed nucleation site, as well as a weak coupling force between the material and the substrate. Additionally, due to the large surface tension of molten glass in the liquid–liquid system, the liquid precursor spreads and diffuses readily, eventually forming a separate interface between the 2D liquid film and the substrate. Without the adsorption process, the diffusion barrier of the pre-spread precursor on the substrate is much lower than in CVD methods. Therefore, the combination of the ultra-low nucleation density and ultrafast growth rate lead to the transformation from polycrystalline MoS_2 to single-crystal MoS_2, achieving the elimination of grain boundaries at an ultra-large scale.

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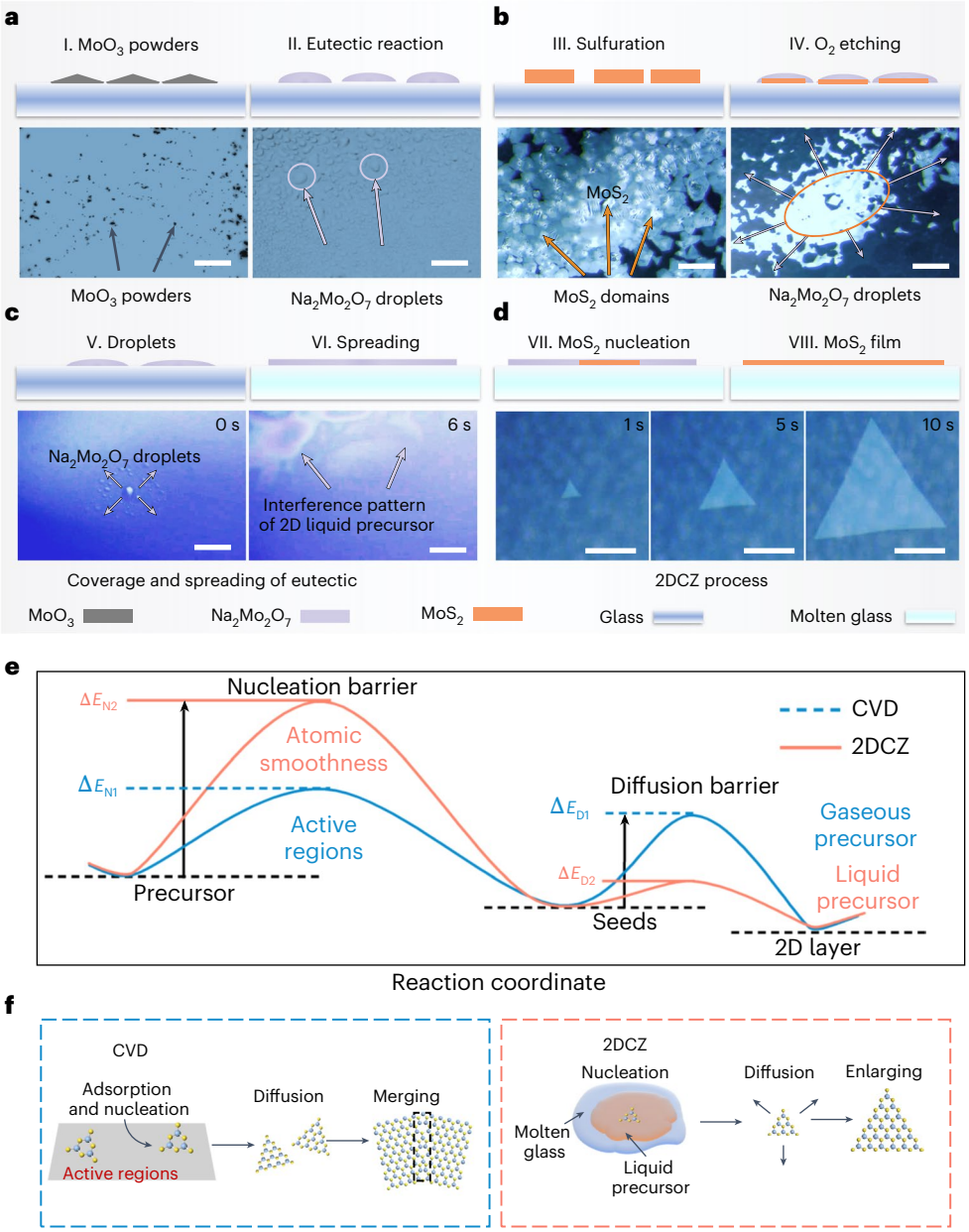

Fig. 2 | T2DCZ mechanism. Schematics (top) and in situ imaging (bottom). **a**, Molten and eutectic reaction of the MoO_3 precursor. Scale bars, 500 µm. **b**, Pre-crystallization and O_2 etching of MoS_2 domains. Scale bars, 500 µm. **c**, Coverage and spreading of the eutectic precursor. **d**, Crystallization from one point to achieve ultrafast large-scale MoS_2 film growth. Scale bars, 300 µm. **e**, Comparison of the progress of energy fluctuation between CVD growth and 2DCZ. Blue lines represent CVD, and red lines represent 2DCZ. **f**, Different interface interactions of conventional CVD, step guide epitaxy and 2DCZ.

Transfer for large-scale MoS_2 film

Because of various constraints, including substrate properties and unsuitability in device construction, transferring nanomaterials is still an essential step in integrating separate modules into a functional system. Using the capillary force and surface tension of deionized water, the MoS_2 film can be peeled off spontaneously (Fig. 3a and Supplementary Video 3). A large-scale MoS_2 film is successfully transferred to a two-inch silicon wafer with high integrity and efficiency (Fig. 3b), while the microscopic surface morphology of the as-transferred MoS_2 (Supplementary Fig. 9) is further studied by atomic force microscopy (AFM) and microscope scanning. This is possible because the characteristics of a smooth surface and weak coupling allow us to adopt the water-assist transferring method, which is significantly different in terms of the chemicals involved, relative to the conventional corrosive method[2,39,40].

For continuous large-area and single-layer 2D materials, the coupling effect of the interface between MoS_2 and the growth substrate is a key to ensuring the overall integrity of transferring[41]. It is the nearly ideal interface status of 2DCZ that enables one to completely separate the MoS_2 film gently from the glass without any etching solution. A chemical etching reaction is needed for a conventional substrate (Fig. 3c). In contrast with a pure aqueous environment, in which the insertion of deionized water offers a mild and effective separation, disruption by gas or boiling leads to a certain extent of damage in the MoS_2 film. As shown in Supplementary Video 4, our transfer process is fully spontaneous: water invades and MoS_2 is separated under capillary action. Note that as-grown MoS_2 without polymethyl methacrylate (PMMA) can still separate spontaneously, similar to a chemical-etching-assisted transfer. Conversely, the external force is necessary in other water-assisted transfer processes[42,43].

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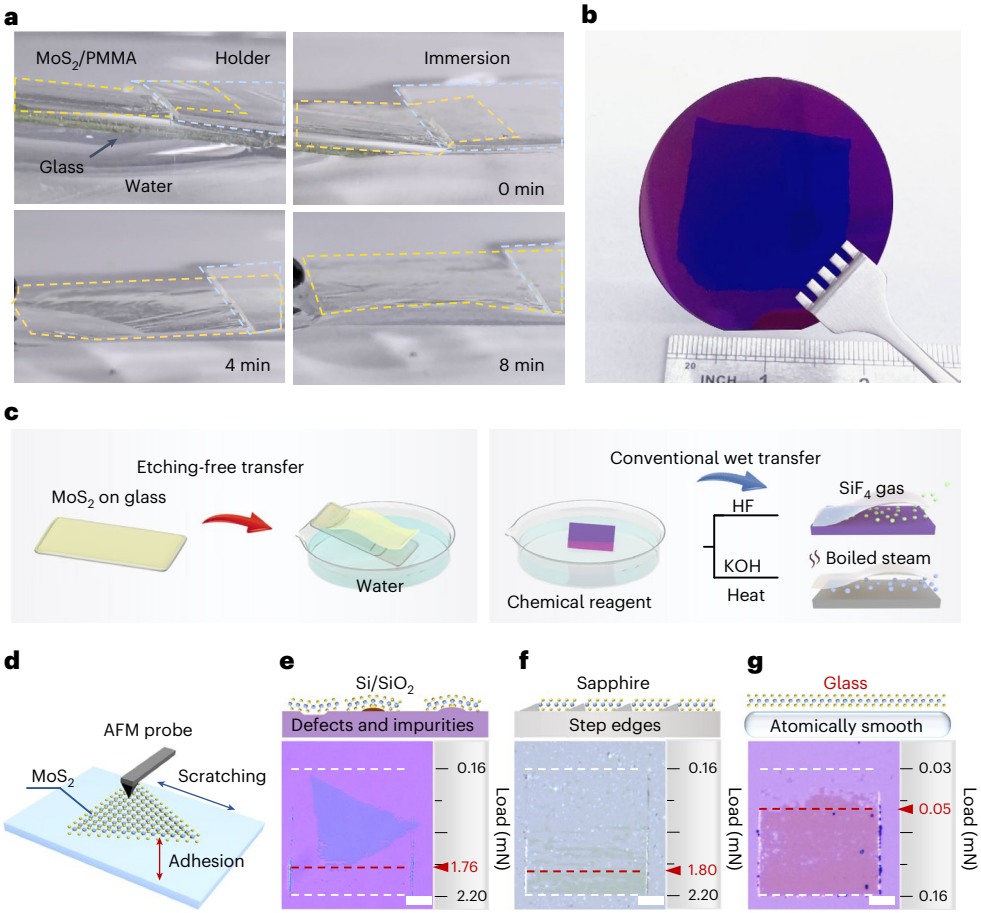

Fig. 3 | Transfer and adhesion between MoS_2 and substrate. a, Photographs of the automated high-efficiency transfer process. **b**, Photograph of the MoS_2 film transferred onto a two-inch silicon wafer. **c**, Comparison with the conventional wet-transfer method. The conventional method requires chemical reagents for transfer, while our MoS_2 can separate spontaneously via water tension in an etching-free environment. **d**, Schematic of the nano-scratch test where the AFM probe moves reciprocatively and the force loaded is proportionally correlated with the voltage applied. **e**–**g**, Results of the nano-scratch test on different substrates; all three mapping areas are 40 × 40 μm², and the inset at the right shows the normalized comparison of load on the surfaces of SiO_2/Si, sapphire and glass, while the top shows a schematic of the surface. Scale bars, 10 μm. The force loaded increases from top to bottom.

To further quantitate the interface adhesion between MoS_2 and various substrates generally used for growth (CVD-MoS_2 grown on SiO_2/Si or sapphire, and 2DCZ-MoS_2 grown on glass), a nano-scratching test was accomplished via AFM (Fig. 3d; more details in the Methods). During the test, the as-grown MoS_2 film on the relative substrate was scraped by the reciprocating motion of the probe along the material surface. The stronger the adhesion between the MoS_2 film and substrates, the more difficult it was to slit the film under the external force. By observing the material morphology, the critical load values for scraping on various growth substrates were determined and compared (Fig. 3e–g). The delamination of MoS_2 on silicon, sapphire and glass substrates was initiated at a load of 1.76 mN, 1.80 mN and 0.05 mN, respectively. The adhesion between the MoS_2 film and glass was the weakest compared with conventional substrates, which is consistent with the energy analysis during growth.

Uniformity and crystallinity at a large scale

After transfer, the large-scale uniformity of the MoS_2 film was assessed through Raman tests. Raman mapping of peak difference (between E^1_{2g} and A_{1g}) on a 2.5 × 2.5 mm² region within the domain area showed minimal variations (Fig. 4a). The average peak difference of 4,000 Raman spectra, statistically analysed, was 18.22 cm^{-1} and followed a Gaussian distribution (Fig. 4b), demonstrating the uniform property of the monolayer MoS_2 film. In principle, the smaller the peak difference between E^1_{2g} and A_{1g}, the lower the defect density of the MoS_2 monolayer[44–46]. Compared with the reported value of 19–20 cm^{-1}, the ultra-low peak difference of 18.22 cm^{-1} suggests the high crystallization quality of the MoS_2 domain[29,43].

The Raman line scan along a centimetre scale exhibited two uniform contour lines perpendicular to the *x* axis (Fig. 4c), further confirming the uniformity of the MoS_2 film over a wide area. The monolayer property of MoS_2 domains is further affirmed through AFM (Supplementary Fig. 1d). Compared with CVD growth, the 2DCZ method effectively confines the growth of MoS_2 to a centimetre-scale 2D space, resulting in uniform monolayer single-crystal growth. Furthermore, nine photoluminescence (PL) mapping patterns of peak intensity across a two-inch MoS_2 wafer (marked area in Fig. 4d) are shown in Fig. 4e. The uniformity of the wafer-scale MoS_2 enhanced the application in large-scale electronic devices.

The continuous crystallization of a uniform precursor ensures the high quality of the as-prepared large-scale MoS_2 domains. To investigate the crystal quality of the MoS_2 film, low-temperature PL was conducted at 83 K (Supplementary Fig. 8a). A sharp full-width at half-maximum of 34 meV with no defect-related emissions demonstrates the high quality of the MoS_2 domain[2]. The single-crystal property of the MoS_2 domain area was further confirmed through low-energy electron diffraction (LEED) tests performed at nine points within a 3 × 3 mm² range (Fig. 4f). The distinct diffraction spots with nearly the same patterns, as expected by C_3 symmetry, affirmed the single-crystal property of MoS_2 in the crystal domain region[7].

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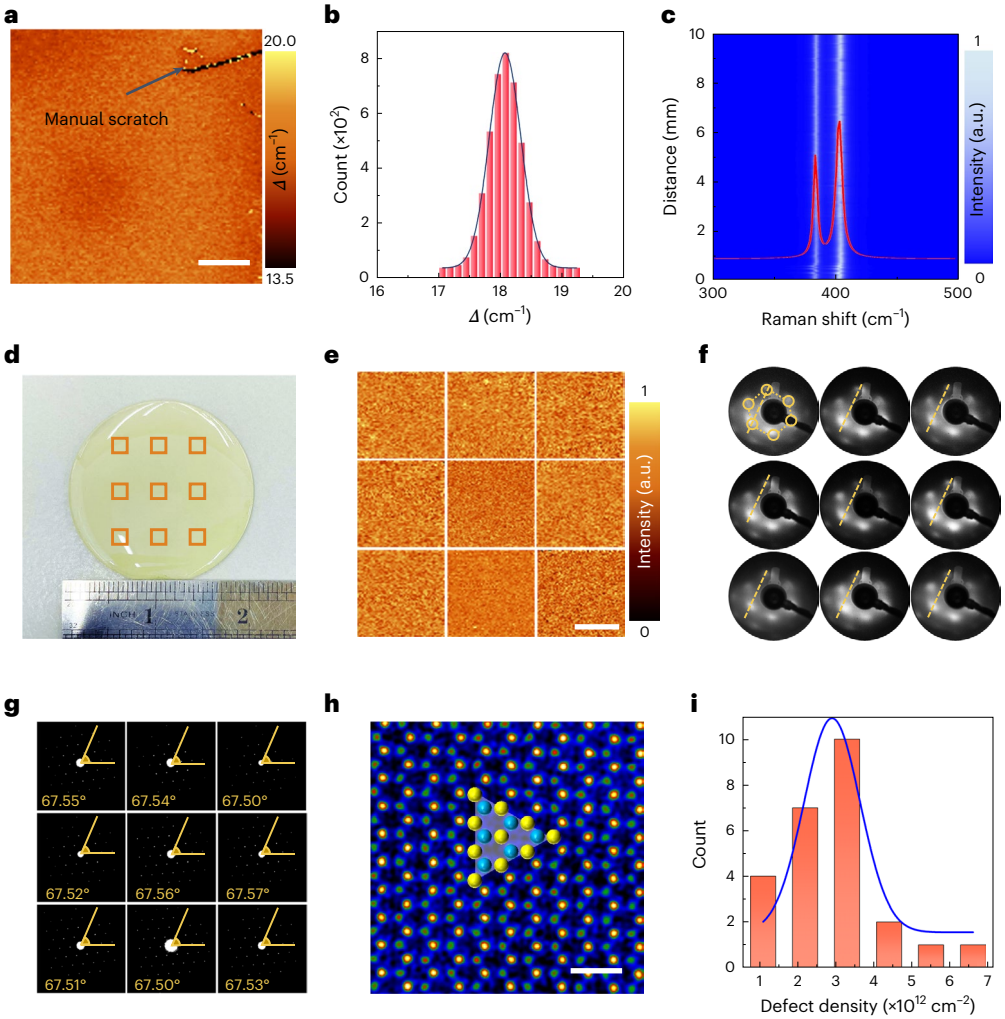

Fig. 4 | Characterization of uniformity and crystal quality of MoS_2. **a**, Raman mapping showing peak difference (between E^1_{2g} and A_{1g}) of the transferred MoS_2 film; the black line scratched by a tweezer is marked for comparison. Scale bar, 500 µm. **b**, Statistical distribution of the Raman peak difference from mapping zones (4,000 Raman spectra). **c**, Raman line scan contour map covering a centimetre length range. **d**, Photograph of a two-inch MoS_2 wafer. **e**, PL mapping patterns of MoS_2 at positions marked in **d**. Scale bar, 10 µm. **f**, LEED patterns of a MoS_2 domain over an area of approximately 6 × 6 mm^2, taken at 80 eV, with the electron beam size around 1 mm. Circles, distinct diffraction spots; dashed lines, auxiliary lines depicting the uniformity of materials. **g**, Selected area electron diffraction of nine points across a 6 × 6 $µm^2$ area. Scale bar, 2 µm. **h**, High-resolution HAADF-STEM image of MoS_2; the inset shows a schematic of the atomic arrangement. Scale bar, 0.5 nm. **i**, Statistical bar graph showing the defect density observed in the HAADF-STEM images. Line, Gaussian fitting curve derived from the defect density statistical data.

The crystal structure and orientation uniformity of the MoS_2 film were examined through aberration-corrected transmission electron microscopy. The selected area electron diffraction was performed at nine points across a 6 × 6 $µm^2$ area. Six-fold symmetric 2H phase patterns of MoS_2 showed almost the same orientation, confirming the single-crystal property[13] (Fig. 4g). In the atomic resolution high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image, the sulfur (blue balls) and Mo (yellow balls) atoms displayed a hexagonal lattice structure arrangement and identical lattice orientations (Fig. 4h). The large-scale HAADF image in Supplementary Fig. 10a further corroborated the excellent uniformity and crystallinity. Furthermore, the sulfur vacancy density was statistically calculated through multi-area HAADF images (Supplementary Fig. 11). The statistical histogram diagram of the sulfur vacancy defect density in 25 different regions within 1 mm on sample with a low variance density of 2.9×10^{12} cm^{-2} (Fig. 4i). This comprehensive characterization confirmed the excellent uniformity and crystallinity of the MoS_2 domain on a large scale. Furthermore, scanning tunnelling microscope (STM) images were obtained, and the high-resolution STM images in Supplementary Fig. 12b–d demonstrate that the STM charge defect density is 6.9×10^{11} cm^{-2}. However, given recent reports on the defect statistics of large-scale, grown MoS_2, measured using STEM, we also adopted STEM statistics to label the sample.

In L–S crystallization, the solute species in the solution diffuse more efficiently, resulting in a stable and uniform precursor for crystal growth. By contrast, the vapour deposition process requires precursor absorption onto and diffusion across the substrate surface, influenced by factors such as gas flow, temperature gradients and the substrate[47]. Furthermore, the low-damage physical transfer process also decreased the defects. Therefore, the high crystalline quality of the as-prepared MoS_2 should be attributed to the L–S crystallization of the 2DCZ process and the low-damage transfer process.

FET performance

The size and quality of our MoS_2 single crystals enabled the fabrication of FET arrays at a centimetre scale[48]. We transferred the MoS_2 films to a prefabricated metal bottom-gate substrate (HfO_2/Au/SiO_2) and fabricated FET arrays using the gate-first process (Fig. 5a,b). Typical transfer

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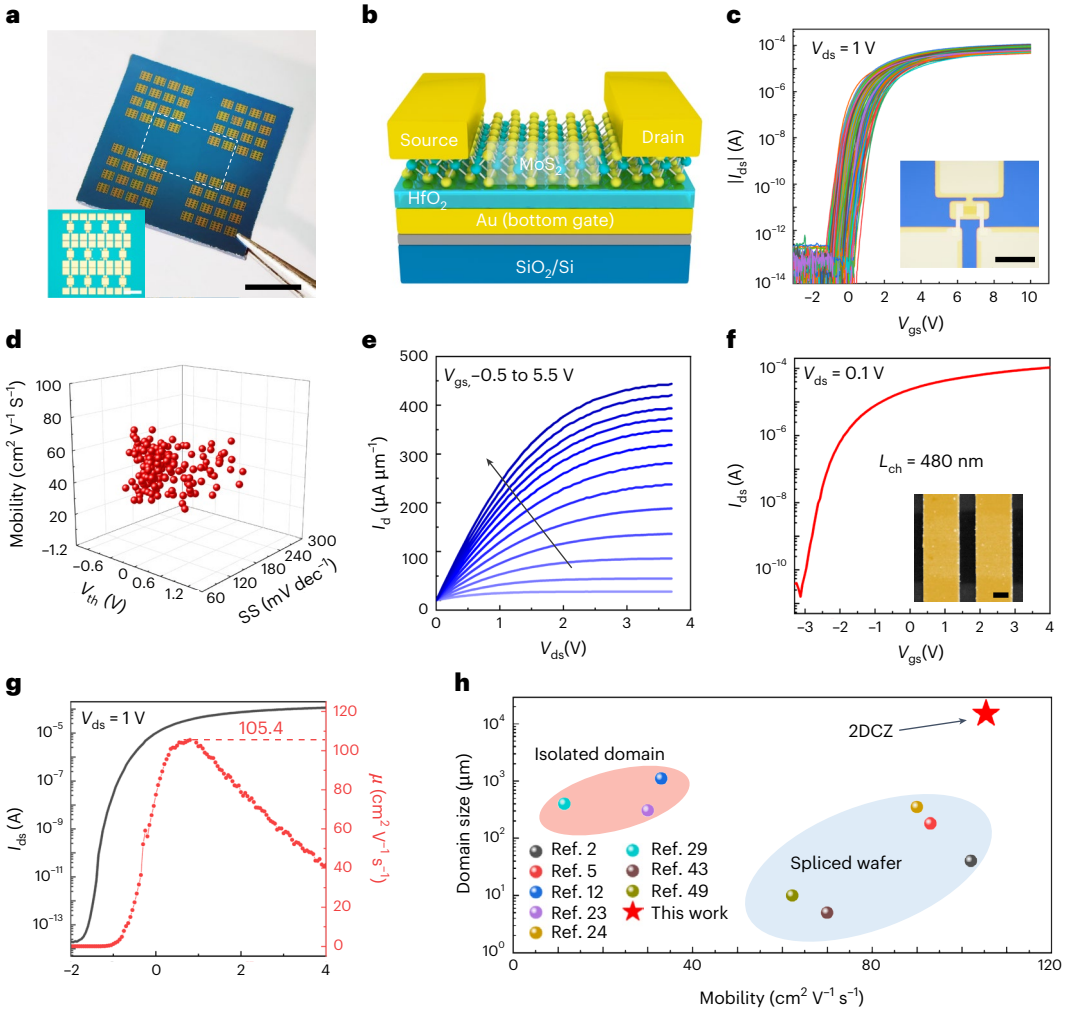

Fig. 5 | Electronic characterizations of the MoS_2 FETs. a, Photograph of a fabricated MoS_2 channel FET array (64 × 24). Scale bar, 5 mm. The inset shows one array unit, which consists of 24 FETs. Scale bar, 200 μm. **b**, Schematic of the MoS_2 FET with Au bottom gate, HfO_2 dielectric and Cr/Au as a contact electrode. **c**, Transfer characteristics of 185 FETs (out of 192) from the marked area in **a**; the inset shows an optical micrograph of a single FET. I_{ds}, drain-source current. Scale bar, 50 μm. **d**, Statistical distribution of SS, mobility and V_{th}, the average values of which are 110 mV dec^{-1}, 55 $cm^2 V^{-1} s^{-1}$ and 0.2 V, respectively. **e**, Output characteristic of the short-channel FET, with V_{gs} from −0.5 to 5.5 V with a 0.5 V step. **f**, Transfer characteristics of the short-channel FET at V_{ds} = 0.1 V. The inset shows a false-coloured scanning electron microscope image of the as-prepared short-channel FET. Scale bar, 500 nm. **g**, Transfer characteristics and mobility of a long-channel FET (L = 16 μm, W = 5 μm) with Bi/Au as the contact electrode on the same substrate. **h**, Benchmark of the 2DCZ growth method. The symbols describe the best mobility and the largest achievable MoS_2 domain sizes in these reports. The red and blue areas mark two growth strategies estimated by the isolated domain and the spliced wafer, respectively.

curves were obtained by testing 192 FETs in the middle of the monolithic area (the marked area in Fig. 5a) and are displayed in Fig. 5c (the inset shows an optical micrograph image of the single FET, with channel length/width L/W = 16/10). Due to the high-quality single-crystal structure, high uniformity of the MoS_2 film and high film transfer integrity, we achieved a device yield of 96.4% (185/192), an advanced level among relative research[2]. The statistical distribution of mobility ($\mu = g_m L/WC_g V_{ds}$, where C_g = 0.47 μF cm^{-2} and where V_{ds}, μ, g_m and C_g are the source-drain voltage, mobility, transconductance and gate capacitance, respectively), subthreshold swing (SS) and threshold voltage (V_{th}) of the FET array are displayed in Fig. 5d and Supplementary Fig. 14. The average mobility of the FET array is 55 $cm^2 V^{-1} s^{-1}$. The variations of mobility, SS and V_{th} are 15.9%, 16.8% and 15.3%, respectively, within the range caused by the damage and pollution introduced by device fabrication[48]. The good uniformity of the device performance is the targeted feature of 2D TMDCs for their 'lab-to-fab' transition[1]. However, the uniformity of device arrays made from 2D materials is currently constrained by processing techniques and remains at a relatively low level[2].

To obtain the saturation current, a short-channel (480 nm) FET device was prepared with Ni/Au as the contact electrode on a 10 nm HfO_2 substrate (C_g = 1.07 μF cm^{-2}; Fig. 5e,f). The device displayed an on-state current of 443.8 μA $μm^{-1}$ at gate voltage (V_{gs}) = 5.5 V (V_{ds} = 3.7 V), a value that might be further improved through the optimization of the electrode evaporation process. Furthermore, the optimal mobility of a single FET is achieved to be 105.4 $cm^2 V^{-1} s^{-1}$ (Fig. 5g). The electrical performance of these devices suggests that the quality of our large-scale monolayer MoS_2 domains is similar to that of exfoliated monolayer MoS_2. Compared with previous work on large-scale MoS_2 growth (including an isolated domain and a spliced wafer), our MoS_2 domain displays a much better domain size and mobility, which indicates the great potential for its integrated application[2,5,12,13,23,24,29,43,49] (Fig. 5h and Supplementary Table 1).

To further demonstrate the applicability of our MoS_2, we engineered integrated logic gate arrays following the pseudo-NMOS logic (Supplementary Fig. 15)[50]. Our inverter comprised two MoS_2 FETs with differing W/L ratios. The voltage transfer curve of the inverter

exhibited a consistent voltage inversion across various supply voltages (V_{dd}) ranging from 0.5 V to 5 V, achieving a peak gain of 171 (V_{dd} = 5 V). Selecting 2 V (about twice switching threshold voltage, ~2V_M) as the optimum V_{dd} for this device, the total noise margin reached an impressive 87%. Additionally, we successfully constructed various basic logic circuits, including NAND, AND, NOR, OR and XOR, with minimal output voltage loss at V_{dd} = 2 V (Supplementary Fig. 16). Due to the excellent uniformity of the 2DCZ-MoS_2 and FET array, our NMOS logic circuit was successfully implemented. The high-quality, highly uniform MoS_2 we prepared significantly promotes the integrated circuit applications of 2D devices.

Conclusion

We have developed a 2DCZ method for the growth of high-uniformity and crystal-quality wafer-level MoS_2 with centimetre-scale domains that will lead to innovations in traditional 2D materials growth methods. Compared with the conventional Czochralski process, the 2DCZ method achieves a 2D liquid precursor on molten glass to suppress vertical crystallization and promote lateral in-plane crystallization. Significantly, the ultrafast crystallization process (75 μm s^{-1}) of MoS_2 from the melted liquid precursors is in situ characterized, which reveals the intricacies of the 2DCZ mechanism. This method, in contrast with the highly random process of conventional CVD, greatly improves the quality, scale and efficiency of the MoS_2 growth. The advanced electronic performance of mobility (105.4 $cm^2 V^{-1} s^{-1}$) and on-state current (443.8 μA $μm^{-1}$) further highlight the superiority of the MoS_2. The 2DCZ method aligns with silicon-based manufacturing processes and offers guidance for the growth of 2D TMDCs, further advancing the industrialization of 2D materials.

Online content

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Methods

2DCZ method of centimetre-scale MoS_2 on molten glass

The growth environment of MoS_2 is a single-temperature zone tube furnace (Thermal, 1,000 mm heating length) at atmospheric pressure. A schematic diagram of the tubular furnace is shown in Supplementary Fig. 1. Excess sulfur powder located upstream of the tubular furnace is heated with a heating belt (135 °C) to provide a sulfur vapour atmosphere (sulfur powder, Aladdin, 99.99%). The glass substrate is supported by a sapphire sheet and placed in the centre of the tube furnace. The Mo source (MoO_3 powder, Aladdin, 99.9%) is located 15 cm upstream of the glass. The high-purity argon (99.9%; flow rate, 200 sccm) is introduced into a 1,500-mm-long, 35-mm-diameter quartz tube to provide an inert atmosphere. The sulfur vapour pressure is controlled by both the temperature and the size of the ceramic crucible holding the sulfur powder. Here the size of the ceramic crucible is 5 cm × 1 cm × 3 cm (length × width × height). The tube furnace is heated to 900 °C at a heating rate of 30 °C min^{-1}, stabilized for 5 min, then heated to 1,100 °C at 7 °C min^{-1}, held for 15 min and then cooled to room temperature. Some O_2 (99%; flow rate, 1.5 sccm) is injected at 1,100 °C for 15 min.

In situ imaging of liquid–solid process

A small amount of MoO_3 powder is dispersed with ethanol onto a sodium–calcium glass surface and placed in the centre of a tubular furnace with a water-cooling system. Argon gas of 20 sccm is fed into the tube furnace as a protection gas and the sulfur vapour is heated to 1,100 °C at a rate of 100 °C min^{-1}. When it is observed that the white crystal on the glass surface stops forming, 0.2 sccm oxygen is injected. When the white crystal is observed to disappear, the oxygen is cut off and the sulfur vapour is reintroduced. When the triangular film stops growing, the experiment ends. At each stage I, II and III, the heating is stopped and the water-cooling system is used to rapidly cool the temperature. The chemical components of the surface of the glass are confirmed by a Raman test in a controlled experiment.

Controlled experiment of eutectic reaction

A small amount of MoO_3 powder is dispersed with ethanol onto a sodium–calcium glass surface and placed in the centre of a tubular furnace with a water-cooling system. Argon gas of 20 sccm is injected into the microtube furnace as a protective gas and heated at a rate of 100 °C min^{-1}. The heating is stopped at 500–1,100 °C, 1,100 °C (under sulfur vapour atmosphere), 1,100 °C (under O_2 vapour atmosphere) and 1,100 °C (under sulfur vapour atmosphere), and the water-cooling system is used to cool the furnace to room temperature within 3 min. The reaction production on the surface of the glass is confirmed by a Raman test.

Adhesion characterization between substrate and MoS_2

The nano-scratch test is obtained using a commercially available AFM instrument (Bruker Dimension ICON system) in air. By applying a voltage to the piezoelectric ceramic of the AFM instrument and inducing deformation, we controlled the contact between the probe and the sample, enabling force to be loaded onto the MoS_2 surface. The loaded force is a product of voltage, deflection sensitivity (7.5 µm V^{-1}, equivalent to half the length of the probe) and k factor (42 N m^{-1}; this constant is related to the probe). In our study, we calibrated the sensitivity and k to maintain constancy during the nano-scratch test. Consequently, voltage serves as an indicator of the force magnitude, with higher applied voltage resulting in a more significant loaded force.

Transfer of wafer-scale monolayer MoS_2

First, 4% PMMA (950 K) is spin-coated on MoS_2/glass at a speed of 1,000 rpm for 30 s and then solidified at a high temperature of 120 °C for 1 min as the polymer supporting layer. This step should be repeated two times to ensure sufficient thickness and strength. Second, a scalpel is used to split the edges of the PMMA/MoS_2/glass to ensure the deionized water can seep into the interface between PMMA/MoS_2 and glass. PMMA/MoS_2 separates from the glass by the action of the water surface tension. After the PMMA/MoS_2 completely separates from the glass (growth substrate) and floats on the deionized water, it can be picked up by the target substrates. Next, the PMMA/target substrate structure is placed at room temperature to remove moisture between the MoS_2 and target substrate interfaces and then baked at 120 °C for 20 min to soften and release the PMMA (polymer supporting layer). Finally, as-transferred samples are soaked in acetone to remove PMMA and then washed in isopropanol and dried by nitrogen gas.

Preparation of devices and electrical measurements

For the FET array and logical device fabrication, the Cr/Au electrode is evaporated on a SiO_2/Si substrate. A high-κ dielectric layer (HfO_2) is deposited on the whole SiO_2/Si substrate. Extra via holes are etched through the dielectric layer by SF_6 reactive ion etching. This is the bottom-gate substrate. A centimetre-scale MoS_2 single crystal is transferred to the as-prepared bottom-gate substrate, assisted by water. The MoS_2 is patterned by electron beam lithography and Ar reactive ion etching. The Cr/Au electrode is used as the contacts, which are defined by electron beam lithography and the evaporation process. The devices are vacuum annealed at 150 °C for 2 h.

For a short-channel FET, a centimetre-scale MoS_2 single crystal is transferred onto a 30 nm substrate of HfO_2 and highly doped Si. The channel areas are isolated by electron beam lithography and Ar etching. Then electron beam lithography is used to define the source/drain contacts. Finally, the devices are coated by thermal vaporization with 10 nm Bi/10 nm Au and lift off is done without an annealing treatment. Electrical measurements are performed using the Agilent B1500A semiconductor tester in a cryogenic probe station (Lakeshore CRX-4K) with a base pressure of ~10^{-5} Pa.

2DCZ method of wafer-scale MoS_2 film on molten glass

The growth environment of MoS_2 is a single-temperature zone tube furnace (Thermal; 1,500 mm heating length) at atmospheric pressure. Excess sulfur powder located upstream of the tubular furnace is heated with a heating belt (145 °C) to provide a sulfur vapour atmosphere (sulfur powder, Aladdin, 99.99%). The glass substrate is supported by a two-inch sapphire and placed in the centre of the tube furnace. The Mo source (MoO_3 powder, Aladdin, 99.9%) is located 10 cm upstream of the glass. The high-purity argon (99.9%; flow rate, 500 sccm) is introduced into a 1,500-mm-long, 700-mm-diameter quartz tube to provide an inert atmosphere. The sulfur vapour pressure is controlled by both the temperature and the size of the ceramic crucible holding the sulfur powder. Here the size of the ceramic crucible is 5 cm ×1 cm × 3 cm (length × width × height). The tube furnace is heated to 900 °C at a heating rate of 30 °C min^{-1}, stabilized for 5 min, then heated to 1,100 °C at 7 °C min^{-1}, held for 30 min and then cooled to room temperature. Some O_2 (99%; flow rate, 1.5 sccm) is injected at 1,100 °C for 30 min.

Characterizations

The PL and Raman spectra are measured in air using a confocal Raman microscopy system (WITec alpha300R, Horiba Jobin Yvon HR800, LabRAM HR Evolution) with an excitation wavelength of 532 nm and a laser spot of 2 µm. Low-temperature PL is measured in a Montana low-temperature stage using Iso Plane SCT-320 with a MGL-FN-532 laser. LEED is employed with a four-grid detector (Omicron Spectra LEED) in a connected ultra-high vacuum chamber to identify the superstructure macroscopically, and the sample is placed in a vacuum chamber and degassed overnight at 270 °C. STEM images are obtained by using the JEOL-ARM200CF operating at 200 keV with double spherical aberration (Cs) correctors. The STM images are obtained with a Nanoscope III scanning probe microscope system (Bruker) under ambient conditions in

constant current mode (inset, 750 pA; voltage bias, 320 mV). A mechanically sharpened Pt/Ir wire was used as a STM tip. AFM is performed using a commercially available AFM instrument (Bruker Dimension ICON system) in air. Electrical transport properties are tested with an Agilent B1500 semiconductor parameter analyser in air. In situ images are taken by a Micro-STS1200 system (Units Technology).

Data availability

All data are available in the main text or Supplementary Information. Source data are provided with this paper.

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Author contributions

H.J., Z.Z. and Y.Z. initiated and supervised the project. H.J. and X.Z. created the MoS_2 monolayers using the 2DCZ method. K.C., X.H. and Y.W. performed device fabrication, data collection and analysis. Y.L., H.Y., L.G. and M.H. assisted in carrying out the mechanism. L.G. performed part of the Raman, PL, LEED and characterization analyses. H.Y. performed the AFM analysis and nano-scratch test. H.J., K.C., X.H., Z.Z., X.Z. and Y.Z. cowrote the paper. All authors discussed the results and commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

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